How was find all formulas of maths

This book belong to the:

A free owl

Index

Note: This library is in developing you can collaborate opening a pull request in `https://github.com/White-Mask-230/Atlas-of-Knowledge-O.-S.-project` or sending me a email with the demonstration of a formula to `lucasvarelacorrea@gmail.com`

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Chapter 1

Initial part

1.1 Introduction

1.1.1 Atlas of Knowledge O. S. project

Goal

Collect all the knowledge of the world in only one site

Differentiation of other projects

- This project is Open Source (that's why the name have O. S.) to protect the books of this project of malicious changes the people that want to collaborate create a pull request that is like a proposal of make a modification that have to be accept for me or it send a email to `lucasvarelacorrea@gmail.com`, it will be accept all except is a modification that is out of topic
- The norms of this project are more soft than other similar projects
- This books can be use as primary source
- This project focus in demonstrated the information know only store it that's why, all things have to be demonstrated or cited
- It is respected the work of other persons that' s why anyone can remove anything that have been create by other person but it can be say that is wrong
- It is allowed for the creators of a paper to put things of his paper in the books of this project
- The project or any book can receive proposals to be change with a explication of why is better the new rules writing a email to `lucasvarelacorrea@gmail.com`

1.1.2 Book

Goal

This book have the goal to collect how was find all the formulas of maths.

Book nomenclature

The Error Zone is where you will find all the formulas that a person find a error in the paper.

Before every formula will be his name and then his id present like this ("chapter-id"."id-chapter-formula") then it can have some comments that the author want to give to the reader:

- The formula is was find in a paper cited in the reference: ["id-reference"]
- The formula is a definition but there is any reference: [x]
- The formula was wrote less than a month: (*)
- The formula was wrote more than a month: (+)
- The formula is cited in the Error Zone: (-)
- Use a formula that is not demonstrated, without any cite and is not a popular formula: (?)

The editor will write clarifications or cited formulas that is in this book like this

- Note: "text-here"
- Note: Use (id "id-formula")

1.2 Error Zone

Chapter 2

Mathematical analysis

2.1 Calculus

2.1.1 Limits

Note: In this subsection there is not how was find the formulas in the next yes (sorry about that)

Properties

If only constant (id 2.1) [1] (x)

$$\lim_{x \to c} k = k$$

If only the term that approximates to (id 2.2) [1] (x)

$$\lim_{x \to c} x = c$$

If there is a constant and a function (id 2.3) [1](x)

$$\lim_{x \to c} k \cdot f(x) = k \lim_{x \to c} f(x)$$

If there are two functions that are $\pm$ (id 2.4) [1] (x)

$$\lim_{x \to c} [f(x) \pm g(x)] = \lim_{x \to c} f(x) \pm \lim_{x \to c} g(x)$$

If there are two functions that multiply (id 2.5) [1] (x)

$$\lim_{x \to c} [f(x) \cdot g(x)] = \lim_{x \to c} f(x) \cdot \lim_{x \to c} g(x)$$

If there are two functions that divide (id 2.6) [1] (x)

$$\lim_{x \to c} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)}$$

If one function composite the other function (id 2.7) [1] (x)

$$\lim_{x \to c} f(g(x)) = f(\lim_{x \to c} g(x))$$

Theorem

Squeeze Theorem $f(x) \leq g(x) \leq h(x)$ (x) [1] (id 2.8)

$$\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L \to \lim_{x \to c} g(x) = L$$

Definition of a derivative (id 2.9) (x)

$$\frac{d}{dx} f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Definition of a integral (id 2.10) (x)

$$\int_a^b f(x)dx = \lim_{n \to \infty} \frac{b-a}{n} \sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right)$$

2.1.2 Summation

Definition summation (id 2.11) (x)

$$\sum_{i=j}^{n} f(i) = f(j) + f(j+1) + \cdots + f(n-1) + f(n)$$

Properties

Summation of the addition or subtraction of two function (2.12) (*)

$$\sum_{i=j}^{n} f(i) \pm g(i) = f(j) + g(j) + f(j+1) + g(j+1) + \cdots + f(n-1) + g(n-1) + f(n) + g(n)$$

$$= f(j) + f(j+1) + \cdots + f(n-1) + f(n) + g(j) + g(j+1) + \cdots + g(n-1) + g(n)$$

Note: Use (id 2.11)

$$= \sum_{i=j}^{n} f(i) \pm \sum_{i=j}^{n} g(i)$$

Summation of a constant and a multiplication (2.13) (*)

$$\sum_{i=j}^{n} C \cdot f(i) = C \cdot f(j) + C \cdot f(j+1) + \cdots + C \cdot f(n-1) + C \cdot f(n)$$

$$= C[f(j) + f(j+1) + \cdots + f(n-1) + f(n)]$$

Note: Use (id 2.11)

$$= C \sum_{i=j}^{n} f(i)$$

Summation of a constant (id 2.14) (*)

Note $C_1 = C_2 = \cdots = C_{n-1} = C_n$ and Use (id 2.11)

$$\sum_{i=1}^{n} C = C_1 + C_2 + \cdots + C_{n-1} + C_n = C \cdot n$$

Theorem

Definition functions or numbers

2.1.3 Derivative

Properties

Derivative of constant multiply function (id 2.15) (*)

$$\frac{d}{dx}[Cf(x)] = \lim_{h \to 0} \frac{C \cdot f(x+h) - C \cdot f(x)}{h}$$

$$= C \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Note: Use

$$= C \frac{d}{dx} f(x)$$

Derivative of the addition or subtraction of two functions (id 2.16) (*)

$$\frac{d}{dx}[f(x) \pm g(x)] = \lim_{h \to 0} \frac{f(x+h) \pm g(x+h) - f(x) - g(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \pm \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$

$$= \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$$

Derivative of a multiply of two functions (id 2.17) (*)

$$\frac{d}{dx}[f(x) \cdot g(x)] = \lim_{h \to 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x)}{h}$$

Note: addition and subtraction of $f(x) \cdot g(x+h)$

$$= \lim_{h \to 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x) + f(x) \cdot g(x+h) - f(x) g(x+h)}{h}$$

$$= \lim_{h \to 0} \frac{[f(x+h) \cdot g(x+h) - f(x) \cdot g(x+h)] + [f(x) \cdot g(x+h) - f(x) \cdot g(x)]}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) \cdot g(x+h) - f(x) \cdot g(x+h)}{h} + \lim_{h \to 0} \frac{f(x) \cdot g(x+h) - f(x) \cdot g(x)}{h}$$

$$= \lim_{h\to 0} \frac{g(x+h)\cdot[f(x+h)-f(x)]}{h} + \lim_{h\to 0} \frac{f(x)\cdot[g(x+h)-g(x)]}{h}$$

$$= \lim_{h\to 0} g(x+h)\cdot \frac{f(x+h)-f(x)}{h} + \lim_{h\to 0} f(x)\cdot \frac{g(x+h)-g(x)}{h}$$

$$= g(x)\cdot \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} + f(x)\cdot \lim_{h\to 0} \frac{g(x+h)-g(x)}{h}$$

$$= g(x)\cdot \frac{d}{dx}[f(x)] + f(x)\cdot \frac{d}{dx}[g(x)]$$

Derivative of a division of two functions (id 2.18) (*)

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{d}{dx}[f(x)\cdot g^{-1}(x)]$$

Note: Using (id 2.17)

$$= g^{-1}(x)\cdot \frac{d}{dx}f(x) + f(x)\cdot \frac{d}{dx}g^{-1}(x)$$

$$= g^{-1}(x)\frac{d}{dx}f(x) - f(x)\cdot g^{-2}(x)\cdot \frac{d}{dx}g(x)$$

$$= \frac{g(x)\frac{d}{dx}f(x) - f(x)\cdot \frac{d}{dx}g(x)}{g^2(x)}$$

Derivative of a constant (id 2.19) (*)

$$\frac{d}{dx}[C] = \lim_{h\to 0} \frac{C-C}{h}$$

$$= 0$$

Derivative of a function elevate by a constant (id 2.20) (*)

$$\frac{d}{dx}f^n(x) = \lim_{h\to 0} \frac{f^n(x+h)-f^n(x)}{h}$$

Note: Using binomial theorem (x)

$$= \lim_{h\to 0} \frac{-f^n(x) + \sum_{k=0}^{n} \binom{n}{k} f^{n-k}(x+h)\cdot h^k}{h}$$

$$= \lim_{h\to 0} \sum_{k=1}^{n} \frac{\binom{n}{k} f^{n-k}(x+h)\cdot h^k}{h}$$

$$= \lim_{h\to 0} \sum_{k=1}^{n} \binom{n}{k} f^{n-k}(x+h)\cdot h^{k-1}$$

Note: Using (id)

$$= \sum_{k=1}^{n} \lim_{h\to 0} \binom{n}{k} f^{n-k}(x+h)\cdot h^{k-1}$$

Note:

$$= \lim_{h\to 0}\left[\binom{n}{1}f^{n-1}(x+h)h^{1-1}+\binom{n}{2}f^{n-2}(x+h)h^{2-1}+\cdots+\binom{n}{k}f^{n-k}(x+h)h^{k-1}\right]$$

$$= \lim_{h\to 0}\binom{n}{1}f^{n-1}(x+h)h^{1-1}+\lim_{h\to 0}\binom{n}{2}f^{n-2}(x+h)h^{2-1}+\cdots+\lim_{h\to 0}\binom{n}{k}f^{n-k}(x+h)h^{k-1}$$

$$= n\cdot f^{n-1}(x)$$

Note: Using the chain rule (x)

$$= n\cdot f^{n-1}(x)\cdot f^{'}(x)$$

Theorem

Definition functions or numbers

2.1.4 Integrals

Properties

Integral of the multiply of a function and a constant (id 2.21) (*)

$$\int_a^b C\cdot f(x)dx = \lim_{n\to\infty}\frac{b-a}{n}\sum_{i=1}^{n}C\cdot f\left(a+i\frac{b-a}{n}\right)$$

$$= C\cdot\lim_{n\to\infty}\frac{b-a}{n}\sum_{i=1}^{n}f\left(a+i\frac{b-a}{n}\right)$$

$$= C\cdot\int_a^b f(x)dx$$

Integral of a addition and a subtraction of two functions (id 1.22) (*)

$$\int_a^b f(x)\pm g(x)dx = \lim_{n\to\infty}\frac{b-a}{n}\left[\sum_{i=1}^{n}f\left(a+i\frac{b-a}{n}\right)\pm g\left(a+i\frac{b-a}{n}\right)\right]$$

$$= \lim_{n\to\infty}\frac{b-a}{n}\left[\sum_{i=1}^{n}f\left(a+i\frac{b-a}{n}\right)\pm\sum_{i=1}^{n}g\left(a+i\frac{b-a}{n}\right)\right]$$

$$= \lim_{n\to\infty}\frac{b-a}{n}\sum_{i=1}^{n}f\left(a+i\frac{b-a}{n}\right)\pm\lim_{n\to\infty}\frac{b-a}{n}\sum_{i=1}^{n}g\left(a+i\frac{b-a}{n}\right)$$

$$= \int_a^b f(x)dx\pm\int_a^b g(x)dx$$

Theorem

Definition functions or numbers

Chapter 3

Final part

3.1 Reference

[1] Alan Stein. Properties of Limits: https://www2.math.uconn.edu/~stein/math115/Slides/math115-130notes.pdf

3.2 Make by

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